Secretary Problem Research

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March 11, 2022

Contents

Lecture 5: Properties of the optimal solution to the relaxed problem with and without error

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1 Simple Problem

We begin with a relaxed problem of best selection with no error. Let $[x, n]$ denote the game state $[x, (x_{n-1}), \ldots, (x_1)]$, i.e., the upcoming item with value x and n un-observed items after the current item.

If the player accepts the first item, he wins iff all other items are less than the first item. Hence $\mathrm{P}[\text{win} \mid \text{accept at } [x, n]] = x^n$. If the player declines x, he wins iff he wins the sub-game $[x_{n-1}, n-1]$, and $\max(x_{n-1}, \ldots, x_1) < x$.

From principle of dynamic programming, the optimal policy can be yielded from the following system of equations:

$$\begin{aligned}
&V(x, 0) = 1 \\
&V(x, n) = \max\left(x^n, (1 - F_{n-1}(x)) E_{x_n} V(x_n, n-1)\right)
\end{aligned}$$

where

$$F_n(x) := \mathrm{P}[\max(x_n, \ldots, x_1) < x \mid [x_n, n-1] \text{wins under optimal policy}].$$

Our first observation is that a thresholding scheme can be optimal.

Theorem 1. There exists a sequence $T_n \subset \mathbb{R}$ such that $x^n > (1 - F_{n-1}(x)) E_{x_n} V(x_n, n-1) \Leftrightarrow x > T_n$ for all $n \in \mathbb{N}$. The same is true if we replace all $>$ with $\geq$.

Proof. The LHS of the inequality monotonely increase wrt x, while the RHS monotonely decrease wrt x. Hence the solution set is in the form $(T_n, 1]$. Specifically, $T_n = \left((1 - F_{n-1}(x)) EV(\cdot, n-1)\right)^{\frac{1}{n}}$. □

We can also say something about T_n:

Theorem 2. The optimal thresholding sequence T_n has the following properties:

1. $T_0 = 0$ and $T_1 = \frac{1}{2}$
2. T_n strictly monotone increase.
3. $\lim_{n\to\infty} T_n = 1$

Proof. TBD. □

Now we examine the asymptotic behavior of the optimal policy.

Theorem 3. The optimal policy ensures a $\frac{1}{2}$ winning rate, and the bound is tight for large n. In other words, $V(0,n) \downarrow \frac{1}{2}$.

Proof. TBD. □

We obtain recipe for the optimal policy by iterated calculation:

$$
\begin{aligned}
EV(\cdot, n) &= \int_0^{T_n} (1 - F_{n-1}(x)) EV(\cdot, n-1)\, dx + \int_{T_n}^1 x^n\, dx \\
F_n(x) &= \frac{\int_0^x \int_0^x \mathbf{1}(z < T_n \wedge z < y) + \mathbf{1}(z > T_n \wedge z > y) F'_{n-1}(y)\, dydz}{\int_0^1 \int_0^1 \mathbf{1}(z < T_n \wedge z < y) + \mathbf{1}(z > T_n \wedge z > y) F'_{n-1}(y)\, dydz} \\
F_0(x) &= x.
\end{aligned}
$$

Numerical result for T_n is summarized in the following table. (Numerical experiments TBD)

2 Relaxed Problem with Error

Fix $\delta > 0$, and use the following notation for game state: $[\tilde{x}, n]$ means the current item to be decided has observed (inaccurate) value $\tilde{x}$, and we have n items after the current item. If the player examines the current item, the game state evolves into $[x, n-1]$, where we now have the exact value x, and one item (the next) is omitted. If the player otherwise declines the current item, the game state evolves into $[\tilde{x}_n, n-1]$ where $\tilde{x}$ is replaced by $\tilde{x}_n$, the next item, and we have one item less in the queue.

Define $X^*_{[x,n]}$ be r.v. representing the value of accepted item from a winning run of the game starting from the specified state, where the policy is assumed to be optimal. We again obtain the DP equations.

$$V(\tilde{x}, 0) = V(x, 0) = 1 \tag{1}$$

$$V(\tilde{x}, n) = \max\left(\mathrm{E}[\mathrm{P}[X_2 < X^*_{[X,n-1]}]V(X, n-1)], \mathrm{E}[\mathrm{P}[X < X^*_{[\tilde{Y},n-1]}]V(\tilde{Y}, n-1)]\right) \tag{2}$$

$$V(x, n) = \max\left(x^n, \mathrm{E}[\mathrm{P}[x < X^*_{[\tilde{Y},n-1]}]V(\tilde{Y}, n-1)]\right). \tag{3}$$

The problem with error is again solvable using thresholding:

Theorem 4. The optimal policy for the problem with error can be described completely by thresholding. In other words, for each $\delta \geq 0$, there exists sequences T_n and $R_n \in \mathbb{R}$ such that the LHS in the max operator of (2) is greater than the RHS iff $\tilde{x} > R_n$. Similarly, the LHS is picked in (3) iff $x > T_n$.

Proof. In (3), note that $\mathrm{P}[x < X^*_{[\tilde{Y},n-1]}]$ is $1-$ a CDF. Taking expectation won't affect monotonicity, so the argument in the previous theorem applies.

Now consider (2). We define $A(x) := \mathrm{P}[X_2 < X^*_{[x,n-1]}]V(x, n-1)$ and $B(x) := \mathrm{E}[\mathrm{P}[x < X^*_{[\tilde{\cdot},n-1]}]V(\tilde{\cdot}, n-1)]$. Our goals is showing that the inequality $EA > EB$, where expectation is taken over $x \in [u-\delta, u+\delta] \cap [0, 1]$, is true iff $u > R_n$ for some $R_n \in \mathbb{R}$. We will first prove that the solution of $A(x) > B(x)$ is a thresholding on x. Then complete the proof by considering properties of expectation.

Let $x = 0$. Now we have $A(0) = \mathrm{P}[X_2 < X^*_{[0,n-1]}]\,\mathrm{P}[0 < X^*_{\tilde{\cdot},n-2]}]V(\tilde{\cdot}, n-2) = \mathrm{P}[X_2 < X^*_{[\tilde{\cdot},n-2]}]V(\tilde{\cdot}, n-2)$, and $B(0) = V(\tilde{\cdot}, n-1)$. $B(0)$ is the probability of winning a game of n items, while $A(0)$ is also the probability of picking the best item from n items, but the exact item X_2 must not be optimum. Hence $A(0)$ is a sub-event of $B(0)$, so $A(0) < B(0)$.

Let $x < T_n$ and now we compute $A(x) = \mathrm{P}[X_2 < X^*_{[x,n-1]}]\,\mathrm{P}[x < X^*_{[\tilde{\cdot},n-2]}]V(\tilde{\cdot}, n-2) = \mathrm{P}[X_2 < X^*_{[\tilde{\cdot},n-2]}]\,\mathrm{P}[x < X^*_{[\tilde{\cdot},n-2]}]V(\tilde{\cdot}, n-2) = A(0)\,\mathrm{P}[x < X^*_{[\tilde{\cdot},n-2]}]$, which monotonely decrease wrt x. $B(x) = B(0)\,\mathrm{P}[x < X^*_{[\tilde{\cdot},n-1]}]$ also monotonely decrease wrt x. $A(x) < B(x)$ follows from the lemma:

Lemma 1. As n increases, the distribution of $X^*_{[0,n]}$ is strictly more right-skewed. To put precisely,

$$\mathrm{P}[x < X^*_{[0,n]}] < \mathrm{P}[x < X^*_{[0,m]}]$$

whenever $n > m$ for all $0 < x < 1$.

Now let $x \geq T_n$. Then $A(x) = \mathrm{P}[X_2 < X^*_{[x,n-1]}]x^{n-1} = x^n$, which monotonely increase up to 1. $B(x)$ still monotonely decrease. We can also verify that $A(x)$

and $B(x)$ are continuous. Thus we can conclude that there exists a unique point $\alpha \in [T_n, 1]$ such that $A(x) > B(x)$ iff $x > \alpha$ for all $x \in [0, 1]$. □

Again we can say something about the thresholding sequences.

Theorem 5. The optimal thresholding sequences T_n and R_n has the following properties:

1. When $\delta = 0$, $T_n \equiv R_n$ are same as the problem without error.
2. For fixed δ, T_n and R_n both strictly monotone increase wrt n.
3. For fixed n, T_n and R_n both monotone decrease wrt δ.
4. $\lim\limits_{n\to\infty} T_n = 1$
5. (?) $\lim\limits_{n\to\infty} R_n = 1 - \delta$.
6. As $\delta \to \infty$, $R_n^{(\delta)} \to -\infty$ and $T_n^{(\delta)} \to T^{(0)}_{\lceil \frac{n}{2} \rceil}$ for all n.

Proof. TBD. □

Corollary 1. The optimal wininng rate $V(0, n)$ satisfies the following:

1. $V_{(0)}$ is identical to the problem without error.
2. As $\delta \to \infty$, $V_{(\delta)}(0, n) \to V_{(0)}(0, \lceil \frac{n}{2} \rceil)$ for all n.

We can compute the optimal thresholds iteratively. Numerical results are summarized below. (Numerical experiments TBD)